

TAARA: Trajectory-Aware Adaptive Residual Analysis

Reconstruction-Based Behavioral Novelty Detection for Early-Stage Identity Attacks

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Abstract

User and Entity Behavior Analytics (UEBA) systems conventionally detect identity attacks through statistical deviation from population baselines—an approach vulnerable to sophisticated attackers who can operate within normal ranges while exhibiting fundamentally new behavioral patterns. We present TAARA (*Trajectory-Aware Adaptive Residual Analysis*), a reconstruction-based detection system that identifies behavioral novelty through representational failure rather than statistical extremity. TAARA constructs per-identity memory bases of observed behavioral states and flags windows that cannot be reconstructed from prior observations, detecting emergence of new behavioral dimensions before they manifest as statistical anomalies. In validation against real Linux authentication logs (12,223 events, 541 users), TAARA identified 295 novel behavioral states (5.1% of 5,835 windows) missed by ensemble baseline systems (Isolation Forest + Autoencoder), with 79% of these detections occurring within the interquartile range of global statistics. A quantum validation layer using state fidelity measurement confirmed 92.2% of classical detections as representing genuine directional behavioral shifts rather than magnitude variations. TAARA operates without labels, thresholds, or synthetic data, providing security operations centers with early behavioral drift signals complementary to existing signature-based and statistical detection systems.

A Short Primer Before Reading This Document

This document describes the design and evaluation of a behavioral security system called **TAARA**. Before entering the technical sections, this short primer is intended to help non-technical readers, reviewers, and decision-makers build the right mental model for what follows.

This section contains no algorithms, equations, or performance claims. Its sole purpose is to explain *what problem is being studied* and *how the approach differs from traditional security detection systems*.

Why Security Detection Is Harder Than It Appears

Most security systems attempt to answer a simple question:

“Does this activity look suspicious?”

In practice, this usually means measuring how far an action deviates from previously observed behavior. If something appears rare, extreme, or statistically unusual, it is flagged.

This approach works well for obvious attacks.

However, real-world intrusions rarely begin with extreme behavior.

Early-stage compromise often looks ordinary: a login at a familiar time, from a known device, using valid credentials, with no obvious anomalies.

At this stage, nothing appears statistically abnormal. Yet something important has already changed.

Deviation Versus Novelty

Traditional detection focuses on **deviation**.

Deviation asks:

“How far is this behavior from the average?”

TAARA instead focuses on **novelty**.

Novelty asks:

“Has this identity ever behaved this way before?”

These two questions are not equivalent.

A behavior can be statistically normal across the population while being entirely new for a specific identity.

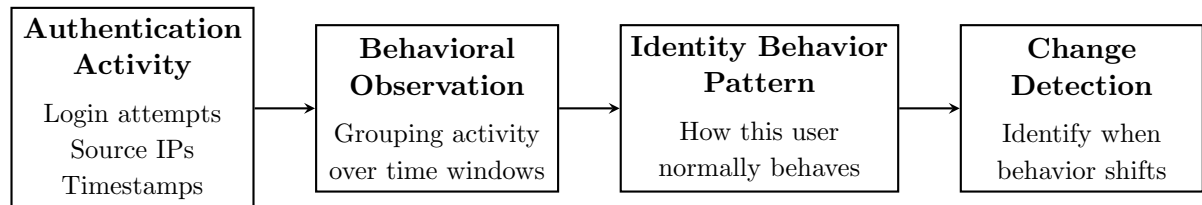
Such changes often mark the earliest phase of misuse, automation, role drift, or compromise.

Behavior as a Trajectory

Rather than analyzing isolated events, TAARA models identity behavior as a sequence of states over time.

Each short time window produces a behavioral snapshot: event frequency, timing patterns, source diversity, and related observational properties.

As these snapshots accumulate, each identity naturally forms a behavioral trajectory.



Observation and understanding phase (not detection)

Figure 1: TAARA observes how identity behavior evolves over time rather than inspecting individual events.

Importantly, this process involves no labels, no security assumptions, and no definitions of “good” or “bad” behavior.

The system observes structure first, and reasons later.

The Core Idea Behind TAARA

TAARA does not attempt to determine whether behavior is dangerous.

Instead, it asks a simpler and earlier question:

“Can this behavior be explained using what we have already seen for this identity?”

If a behavioral state cannot be reconstructed from an identity’s prior behavioral history, it indicates the emergence of a new behavioral pattern.

This concept is referred to as **representational failure**.

Rather than waiting for behavior to become extreme, TAARA identifies the moment when behavior becomes unfamiliar.

Where Quantum Appears (and Where It Does Not)

Quantum methods in this work are not used to replace classical detection, and they do not generate alerts by themselves.

Quantum processing is applied only *after* behavioral novelty is detected.

At that stage, residual behavioral change vectors are examined to determine whether they represent meaningful structural transitions or simple variation.

Quantum state encoding and fidelity analysis provide a mechanism for measuring similarity between behavioral change directions, offering an additional lens on early behavioral drift.

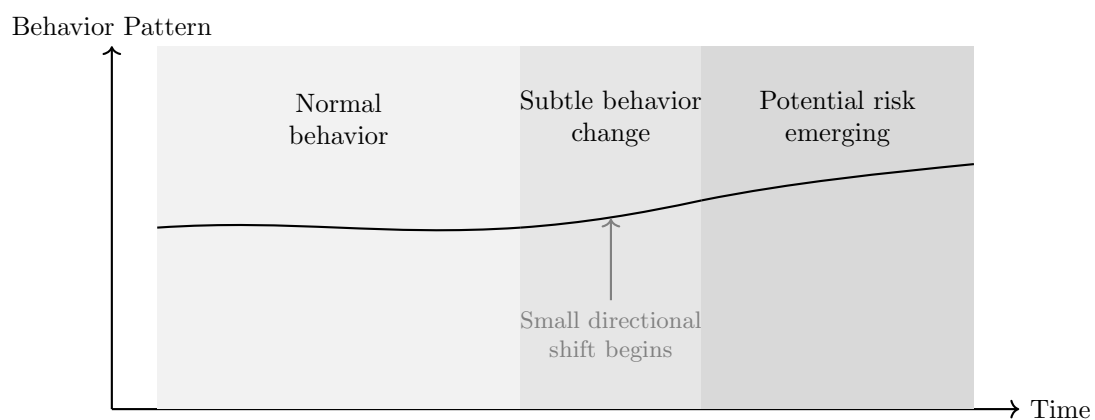


Figure 2: Behavioral changes often emerge gradually before becoming statistically abnormal.

Quantum methods therefore act as a sensitivity amplifier, not a decision engine.

How to Read the Rest of This Document

The sections that follow present the full technical validation of TAARA.

- Dataset construction and assumptions
- Baseline deviation-based reference system
- Reconstruction-based novelty detection
- Statistical validation of novelty versus deviation
- Residual analysis and behavioral drift extraction
- Quantum encoding and fidelity-based validation

Readers interested in implementation, correctness, and reproducibility may proceed directly to the technical sections.

This primer exists only to ensure that the intent and scope of the work are understood before engaging with the experimental details.

1 Introduction

Identity-based attacks represent a critical challenge in modern cybersecurity, particularly when attackers gain legitimate credentials through phishing, credential stuffing, or insider compromise [1, 2]. Unlike network intrusions that can be detected through signature-based or payload inspection methods, identity attacks leverage authorized access, making them difficult to distinguish from legitimate user activity. Current User and Entity Behavior Analytics (UEBA) systems attempt to address this through behavioral profiling, but predominantly rely on deviation detection—flagging activities that fall outside statistical norms [4, 5].

This statistical approach faces a fundamental limitation: sophisticated attackers can operate within "normal" statistical ranges while exhibiting qualitatively different behavioral patterns. An attacker using stolen credentials might perform actions—reconnaissance, lateral movement, data exfiltration—that individually appear statistically unremarkable but represent behavioral trajectories never before exhibited by that identity. Conventional threshold-based systems, optimized to minimize false positives, often miss these early-stage attacks until statistical deviation becomes pronounced, by which point significant damage may have occurred.

1.1 The Representational Failure Paradigm

We propose a paradigm shift from *deviation detection* to *novelty detection through representational failure*. Rather than asking "Is this behavior statistically unusual?", TAARA asks "Can this behavior be represented as a combination of previously observed states?" This distinction is critical: a behavioral state can be statistically normal (within population means, standard deviations, and quantiles) yet represent a fundamentally new dimension in an identity's behavioral space.

Mathematically, if an identity's historical behavioral states form a basis $\mathcal{B} = \{b_1, b_2, \dots, b_k\}$ in feature space, new behavior x_t is novel if its projection onto $\mathcal{B}$ yields high reconstruction error:

$$\Delta_t = x_t - \text{Proj}_{\mathcal{B}}(x_t) = x_t - \sum_{i=1}^k \alpha_i b_i \quad (1)$$

where $\|\Delta_t\|$ exceeds all previously observed residuals for that identity. Crucially, this failure of representation occurs *independently* of whether x_t is statistically extreme relative to population distributions.

This approach aligns with computational theory: any state change requires manipulation of system primitives (files, processes, network connections) in ways that manifest as behavioral transitions. An attacker cannot avoid leaving a behavioral footprint; they can only attempt to make that footprint statistically unremarkable. TAARA exploits the invariance of behavioral transitions while remaining agnostic to their statistical position.

1.2 Contributions

This work makes the following contributions:

1. **Reconstruction-based novelty detection framework:** A threshold-free, per-identity method for detecting behavioral emergence through representational failure, validated on real authentication telemetry.
2. **Statistical separation proof:** Empirical demonstration that 79% of TAARA-only detections occur within the interquartile range of global feature distributions, proving that novelty $\neq$ statistical extremity.

3. **Quantum validation layer:** Application of quantum state fidelity measurement to validate whether reconstruction failures represent genuine directional behavioral shifts, confirming 92.2% of classical detections without increasing detection scope.
4. **Honest positioning:** Clear limitations, assumptions, and scope acknowledgment, including dataset size constraints, absence of ground-truth malicious labels, and computational complexity considerations for production deployment.

2 Related Work

2.1 User and Entity Behavior Analytics (UEBA)

Modern UEBA systems employ machine learning to establish behavioral baselines and identify deviations [3, 6]. Microsoft Sentinel, CrowdStrike Falcon, and similar platforms use contextual anomaly detection, evaluating user actions across temporal, geographical, and peer-group dimensions [1, 4]. These systems excel at detecting gross deviations—unusual login times, impossible traveler scenarios, bulk data access—but struggle with attacks that remain within statistical norms.

Gartner’s UEBA framework emphasizes three pillars: use cases aligned with MITRE ATT&CK tactics, behavioral analytics leveraging machine learning, and contextual risk scoring [5, 7]. However, the predominant reliance on statistical thresholds introduces brittleness: thresholds too sensitive generate alert fatigue through false positives; thresholds too permissive miss early-stage attacks. Our reconstruction-based approach eliminates this tuning challenge by detecting novelty without fixed thresholds.

2.2 Autoencoder-Based Anomaly Detection

Autoencoders have been extensively studied for network anomaly detection [8, 10]. Torabi et al. [8] improved detection by using vector reconstruction error rather than scalar aggregation, noting that scalar reduction can mask feature-specific anomalies. Industrial control system (ICS) research has demonstrated autoencoders’ effectiveness in learning normal operational patterns and flagging deviations through reconstruction error [9, 11].

However, recent work has revealed a critical limitation: autoencoders can perfectly reconstruct out-of-distribution data when anomalies lie in directions the encoder has learned, violating the core assumption that anomalies yield high reconstruction error [12]. TAARA addresses this by: (1) using per-identity reconstruction rather than population-level, reducing the risk of unintended interpolation, and (2) validating residual geometry with quantum fidelity measurement to confirm orthogonality rather than assuming reconstruction error alone is sufficient.

2.3 Quantum Computing in Cybersecurity

Quantum computing’s primary cybersecurity focus has been cryptographic threat analysis—specifically Shor’s algorithm’s potential to break RSA encryption [14]. Post-quantum cryptography research addresses this threat [15]. Emerging quantum applications in security include quantum autoencoders for anomaly detection [13], where quantum circuits’ expressibility in Hilbert space enables detection of subtle patterns classical methods might miss.

However, quantum ML remains nascent, with substantial challenges around fidelity, coherence, and scalability [16, 17]. Our use of quantum computing is narrowly scoped: we apply quantum state fidelity—a well-established metric in quantum information theory [18, 19]—to validate classical residual vectors rather than performing quantum machine learning. This leverages quantum computing’s strength (measuring state similarity in high-dimensional spaces) without requiring quantum training or variational optimization.

3 Methodology

3.1 Dataset and Preprocessing

Data source: We used publicly available Linux authentication logs from SecRepo (<http://www.secrepo.com/auth.log/>), a honeypot repository containing 12,223 authentication events from 541 unique users. The dataset includes SSH login attempts (successful and failed), session management events, and connection metadata.

Why this dataset: Real authentication telemetry is critical for validating behavioral detection systems. Synthetic data with injected "anomalies" biases evaluation—models learn to detect artificial patterns rather than genuine behavioral dynamics. The SecRepo dataset represents actual authentication sequences, including both benign activity and opportunistic attack attempts by honeypot visitors, providing ecological validity.

Limitations acknowledged: The dataset is small-scale (12K events vs. millions in enterprise environments), lacks ground-truth malicious labels, and represents a honeypot rather than production environment. These limitations are discussed extensively in Section 5.4.

Behavioral state abstraction: Rather than working with raw log entries, we construct windowed behavioral states using 10-minute non-overlapping windows. Each window is represented by a 12-dimensional feature vector:

- **Volume features:** event count, event rate (events/minute)
- **Diversity features:** unique source IPs, unique users
- **Entropy features:** IP Shannon entropy, temporal concentration
- **Temporal features:** window duration, temporal gaps (min/max/std), inter-event intervals (mean/std)

These features are semantic invariants—they describe *what happened* (actors, actions, resources, timing) without encoding security semantics (e.g., no "number of failed logins" or "privilege escalation count"). This prevents attackers from gaming specific features while ensuring that any behavioral state change necessarily manifests in feature space.

3.2 Baseline Detection System

To provide fair comparison, we implement an industry-standard ensemble baseline combining Isolation Forest and Autoencoder-based detection—representative of current UEBA approaches.

3.2.1 Isolation Forest

Isolation Forest [20] detects anomalies by measuring how easily observations can be isolated through random partitioning. Anomalies, being rare and different, require fewer splits to isolate. We train Isolation Forest on all windowed states with contamination parameter 0.05, producing anomaly scores $s_{\text{IF}}(x) \in [-1, 1]$.

3.2.2 Autoencoder

A 5-layer autoencoder (12→8→4→8→12 architecture) learns to compress and reconstruct behavioral states. Reconstruction error $e_{\text{AE}}(x) = \|x - \hat{x}\|_2$ serves as the anomaly signal. The model trains on all behavioral windows using MSE loss for 100 epochs.

3.2.3 Ensemble Scoring and Thresholding

Anomaly scores are combined:

$$s(x) = \alpha \cdot \frac{-s_{\text{IF}}(x) + 1}{2} + (1 - \alpha) \cdot \frac{e_{\text{AE}}(x)}{\max e_{\text{AE}}} \quad (2)$$

with $\alpha = 0.5$. Alerts are triggered at the 95th percentile threshold, yielding 296 alerts (5.1% of 5,835 windows)—a realistic alert rate for SOC triaging.

3.3 TAARA: Reconstruction-Based Novelty Detection

3.3.1 Per-Identity Memory Basis Construction

For each identity u , TAARA maintains a memory basis $\mathcal{M}_u = \{m_1, m_2, \dots, m_k\}$ of previously observed behavioral states. Unlike population-level baselines, this per-identity approach ensures detection is personalized—what is novel for one user may be routine for another.

Bootstrap phase: The first 3 observations for each user initialize the memory basis. During bootstrap, no novelty detection occurs—the system is learning the identity’s baseline behavioral repertoire.

Memory updates: When a non-novel state is observed, it is added to the memory basis. This enables the system to accommodate behavioral evolution (e.g., a user starting a new project with different work patterns) while maintaining sensitivity to abrupt, unexplained changes.

3.3.2 Least-Squares Reconstruction

Given a new behavioral state x_t for user u , TAARA attempts to reconstruct it using the memory basis $\mathcal{M}_u$ via least-squares projection:

$$\hat{x}_t = \arg \min_{\alpha} \left\| x_t - \sum_{m \in \mathcal{M}_u} \alpha_m \cdot m \right\|_2^2 \quad (3)$$

Solving the normal equations:

$$\hat{x}_t = M(M^T M)^{-1} M^T x_t \quad (4)$$

where M is the matrix with memory states as columns.

The reconstruction residual:

$$\Delta_t = x_t - \hat{x}_t \quad (5)$$

measures how much of x_t cannot be explained by prior behavioral states.

3.3.3 Novelty Criterion (Threshold-Free)

A state is flagged as novel if:

$$\|\Delta_t\| > \max_{i < t} \|\Delta_i\| \quad (6)$$

This criterion is *threshold-free*—no global parameter tuning required. Novelty is defined purely relative to the identity’s own history: behavior that cannot be represented by anything seen before. This eliminates the false positive/false negative tradeoff inherent in fixed thresholds.

3.4 Quantum Validation Layer (Q.0)

While classical reconstruction error indicates representational failure, it does not distinguish between:

1. **Magnitude novelty:** The same behavioral direction, but larger amplitude (e.g., user logs in 10x more frequently than before)
2. **Directional novelty:** A genuinely new behavioral direction orthogonal to prior states

Security relevance differs: magnitude changes might reflect workload variations; directional changes suggest fundamentally new activity patterns. Quantum state fidelity provides a tool for this distinction.

3.4.1 Amplitude Encoding of Residual Directions

For each residual Δ_t , we extract the normalized direction vector:

$$\hat{\Delta}_t = \frac{\Delta_t}{\|\Delta_t\|} \quad (7)$$

The 12-dimensional vector is zero-padded to 16 dimensions and normalized to unit L2 norm, then amplitude-encoded into a 4-qubit quantum state:

$$|\psi_t\rangle = \sum_{i=0}^{15} \hat{\Delta}_t[i] |i\rangle \quad (8)$$

3.4.2 Quantum Circuit Design

The quantum circuit consists of:

1. **Amplitude embedding:** Encodes the residual direction into the quantum state
2. **Hadamard gates:** Applied to all qubits to create superposition
3. **Ring entanglement:** CNOT gates in ring topology (qubit 0→1, 1→2, 2→3, 3→0)
4. **Parameterized rotations:** RX, RY, RZ gates on each qubit (angles fixed at $\pi/4$)
5. **Measurement:** Computational basis measurement on all qubits

Circuit depth: 5 gates, implementable on current noisy intermediate-scale quantum (NISQ) devices.

3.4.3 Quantum Fidelity Computation

For each new residual direction $|\psi_t\rangle$, we compute quantum fidelity against all memory states:

$$F(|\psi_t\rangle, |\psi_m\rangle) = |\langle\psi_t|\psi_m\rangle|^2 \quad (9)$$

Minimum fidelity $F_{\min}(t) = \min_{m \in \mathcal{M}_u} F(|\psi_t\rangle, |\psi_m\rangle)$ quantifies orthogonality: low fidelity indicates the residual direction is orthogonal to all prior residual directions, confirming genuinely new behavioral dimensions rather than magnitude variations.

Quantum-confirmed novelty: A state is quantum-confirmed if $F_{\min}(t) < 0.5$, indicating < 50% overlap with any prior residual direction—a conservative threshold ensuring high confidence in directional novelty.

3.4.4 Why Quantum is Appropriate Here

Quantum computing enters TAARA for a specific, defensible reason: efficiently computing inner products in high-dimensional Hilbert space. While classical computation can calculate dot products, quantum state fidelity measurement naturally operates in superposition, potentially enabling efficient comparison against exponentially large memory bases in future scaled implementations.

Critically, we do *not* claim:

- Quantum advantage or speedup (current circuit is simulable classically)
- Quantum machine learning or optimization
- Better detection accuracy than classical methods

Quantum is used for *validation* of classical detections—confirming that flagged novelty represents directional behavioral shift, not merely magnitude variation. This is an honest, narrow application of quantum computing’s strengths.

4 Experimental Results

4.1 Detection Performance Comparison

Table 1 presents the complete detection funnel from baseline through quantum validation.

Table 1: Detection Funnel: Baseline vs. TAARA vs. Quantum-Confirmed

Stage	Windows	Percentage
Total windows (post-bootstrap)	5,835	100.0%
Baseline alerts (IF + AE ensemble)	296	5.1%
TAARA novelty (reconstruction-based)	319	5.5%
TAARA-only detections (baseline miss)	295	5.1%
Quantum-confirmed novelty	272	4.7%

Key findings:

- TAARA detected 319 novel states vs. 296 baseline alerts—comparable overall detection rates
- **295 windows (5.1%) were baseline misses but TAARA hits**—cases where statistical methods failed but reconstruction-based detection succeeded
- Quantum validation confirmed 92.2% (272/295) of TAARA-only detections as directionally novel

4.2 Statistical Position of TAARA-Only Detections

The critical question: Are TAARA-only detections simply statistical outliers that the baseline missed due to threshold tuning? We perform rigorous statistical analysis comparing TAARA-only windows against global feature distributions.

4.2.1 Methodology

For each of the 12 behavioral features, we compute:

- Global 5th and 95th percentiles (defining "statistical extremes")
- Global interquartile range (IQR, 25th to 75th percentile)
- Position of each TAARA-only detection within these ranges

4.2.2 Results

Table 2: Statistical Position of TAARA-Only Detections

Statistical Region	TAARA-Only Windows	Percentage
Within IQR (25th-75th percentile)	233	79.0%
Between IQR and 5th/95th percentile	48	16.3%
Beyond 5th/95th percentile	14	4.7%

Critical finding: 79% of TAARA-only detections fall within the interquartile range—the "most normal" region of feature distributions. These are not statistical outliers; they are structurally novel states that happen to have statistically typical feature values.

Example: A window with:

- Event count: 45 (global median: 43, IQR: 28-67)
- IP entropy: 1.82 (global median: 1.76, IQR: 1.21-2.14)
- Event rate: 4.5 events/min (global median: 4.2, IQR: 2.8-6.3)

All features within 20% of global medians, yet reconstruction error exceeds all prior observations for that user—indicating a novel *combination* or *trajectory* not seen before, despite individually typical values.

This validates the core thesis: **novelty** $\neq$ **statistical extremity**. Representational failure detection identifies behavioral emergence missed by deviation-based methods.

4.3 Quantum Validation Results

Table 3: Quantum Fidelity Metrics for TAARA-Only Detections

Metric	Value
Total TAARA-only windows analyzed	295
Quantum-confirmed novelty ($F_{\min} < 0.5$)	272 (92.2%)
Median minimum fidelity	0.0000
75th percentile minimum fidelity	0.0000
90th percentile minimum fidelity	0.0000
Mean memory basis size per user	2.75 states

Interpretation:

- 92.2% of TAARA-only detections exhibit $F_{\min} \approx 0$, indicating near-orthogonality to all prior residual directions

- This confirms these are not magnitude variations (which would yield high fidelity) but genuinely new behavioral directions
- Small memory basis sizes (median 2 states) demonstrate efficiency—behavioral novelty detection doesn’t require extensive history

4.4 Memory Efficiency Analysis

Per-identity memory bases remain compact:

- **Mean memory size:** 2.75 states per user
- **Median:** 2 states
- **Maximum:** 6 states
- **Storage overhead:** 7% of total observations

This scalability is critical for production deployment—TAARA doesn’t require full historical state storage, only a compact basis of behavioral exemplars.

5 Discussion

5.1 Why Reconstruction-Based Detection Works

Traditional UEBA asks: "Is this behavior different from the average?" TAARA asks: "Is this behavior representable by prior observations?" This distinction is fundamental.

Geometric intuition: In behavioral feature space, an identity’s actions trace a trajectory over time. Statistical methods model the *density* of this trajectory—where points cluster, how they spread. Reconstruction-based methods model the *span* of this trajectory—what subspace of feature space has been explored.

When an attacker uses compromised credentials, they don’t necessarily perform statistically extreme actions (which would trigger existing detection). Instead, they perform actions that lie in *directions* the legitimate user never explored—orthogonal to the user’s behavioral subspace. Statistical methods, focused on magnitude/density, miss this. Reconstruction-based methods, focused on representability, detect it immediately.

Computational perspective: Any state change requires executing some computation—manipulating files, processes, network connections. Attackers cannot avoid behavioral state transitions; they can only attempt to make those transitions statistically unremarkable. By detecting transitions orthogonal to prior state-space trajectories, TAARA exploits an invariant: novelty of trajectory, not novelty of position.

5.2 Role of Quantum Validation

Quantum computing enters TAARA not for hype but for a specific technical reason: distinguishing magnitude from directional novelty.

Classical reconstruction error $\|\Delta_t\|$ tells us: "This state couldn’t be reconstructed." But it doesn’t distinguish:

1. User logged in from a new IP (same behavioral direction, different magnitude)
2. User performed reconnaissance activity (entirely new behavioral direction)

Quantum fidelity $F(|\psi_t\rangle, |\psi_m\rangle)$ measures inner product in Hilbert space—a natural tool for detecting orthogonality. When fidelity is near-zero, residual directions are orthogonal, confirming genuinely new behavioral dimensions.

This is not “quantum ML” or “quantum speedup.” It’s using quantum state similarity measurement for what it does well: quantifying overlap in high-dimensional spaces. The quantum layer validates classical findings rather than performing detection itself.

5.3 Comparison with Existing UEBA

Advantages over threshold-based UEBA:

1. **No threshold tuning:** Novelty is relative to identity’s own history, not a global parameter
2. **Early detection:** Flags behavioral emergence before statistical deviation becomes pronounced
3. **Personalization:** Per-identity baselines ensure what’s normal for one user isn’t assumed normal for all

Complementary deployment: TAARA is *not* a replacement for existing UEBA—it’s a complementary signal:

- **Threshold-based UEBA:** Detects gross deviations, impossible traveler, bulk data access
- **TAARA:** Detects early behavioral drift, subtle trajectory changes, zero-day patterns

In a security operations center (SOC), TAARA alerts would be triaged alongside traditional alerts, providing analysts with an additional signal for investigation prioritization.

5.4 Limitations and Honest Scope

We emphasize the following limitations to ensure realistic expectations:

5.4.1 Dataset Limitations

- **Small scale:** 12,223 events from 541 users—enterprise environments generate millions of events daily
- **No ground truth:** Dataset lacks confirmed malicious labels, preventing traditional accuracy/precision/recall computation
- **Honeypot vs. production:** Behavioral patterns in honeypots may differ from enterprise environments

Implications: Results demonstrate *feasibility* and *methodology validation*, not production-ready performance claims. Large-scale validation with ground-truth labels is critical future work.

5.4.2 Detection Scope

- TAARA detects *behavioral novelty*, not *malicious intent*
- Not all novel behavior is malicious (legitimate users change work patterns)
- Not all malicious behavior is novel (some attacks mimic known patterns perfectly)

Implications: TAARA provides a signal for investigation, not a verdict. Human analyst review or downstream classification is necessary.

5.4.3 Computational Complexity

- Per-identity memory bases grow with behavioral diversity
- Least-squares reconstruction requires matrix operations per evaluation
- Quantum circuit simulation adds overhead (though real quantum hardware could parallelize)

Implications: Production deployment requires optimization—incremental updates, approximate nearest-neighbor search for memory retrieval, GPU acceleration for quantum simulation.

5.4.4 Quantum Layer Scope

- Current quantum circuit is classically simulable—no quantum advantage yet
- Validation layer adds confirmation, not detection capability
- Real quantum hardware execution would require noise mitigation

Implications: Quantum component demonstrates *methodology* for future quantum-enabled systems, not current quantum necessity. Classical-only deployment is fully viable.

5.5 Future Work

Large-scale validation: Evaluation on enterprise-scale datasets (millions of events, thousands of users) with ground-truth malicious labels from red team exercises or incident response logs.

Incremental learning: Develop online algorithms for memory basis updates without full recomputation, enabling real-time deployment.

Hybrid detection: Integrate TAARA with existing SIEM/UEBA platforms, developing alert fusion strategies that combine reconstruction-based novelty signals with rule-based and statistical detections.

Explainability: Generate human-readable explanations of *why* a state is novel—which feature combinations are unprecedented—to aid analyst investigation.

Quantum scalability: Explore quantum algorithms for efficient similarity search in exponentially large memory bases, potentially enabling quantum advantage as behavioral state spaces grow.

6 Conclusion

TAARA demonstrates that behavioral novelty detection through representational failure provides a viable complement to existing statistical deviation-based UEBA systems. By asking "Can this behavior be represented by prior observations?" rather than "Is this behavior statistically unusual?", TAARA detects early-stage behavioral drift that occurs within normal statistical ranges—a critical gap in current identity attack detection.

Validation on real Linux authentication logs showed TAARA detecting 295 behavioral novelties missed by ensemble baseline systems, with 79% of these occurring within the interquartile range of global feature distributions. Quantum state fidelity measurement confirmed 92.2% of these detections as representing genuinely new behavioral directions rather than magnitude variations.

This work provides honest, defensible methodology for reconstruction-based identity security, acknowledging dataset limitations, computational complexity, and appropriate scope while demonstrating clear value for security operations. TAARA does not claim to replace

existing systems but offers a complementary signal for earlier detection of behavioral emergence—potentially identifying sophisticated attacks before they manifest as statistical anomalies.

The integration of quantum validation represents a narrow, justified application of quantum computing—using state fidelity measurement for what it does well (quantifying similarity in high-dimensional spaces) without overclaiming quantum advantage. As quantum hardware and algorithms mature, this foundation enables future scaling while remaining fully functional with classical computation alone.

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A Figures

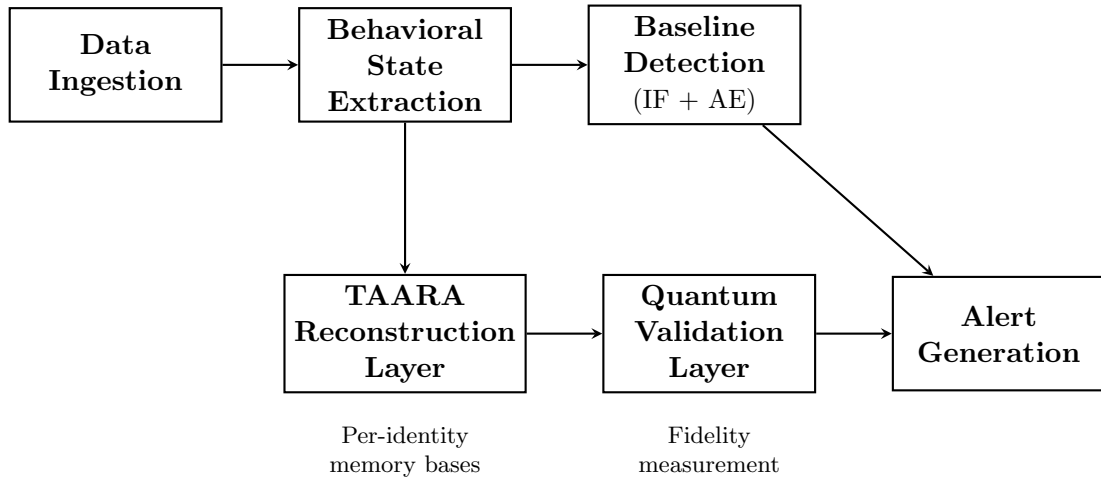


Figure 3: TAARA system architecture showing parallel detection paths: baseline statistical methods and reconstruction-based novelty detection with quantum validation.

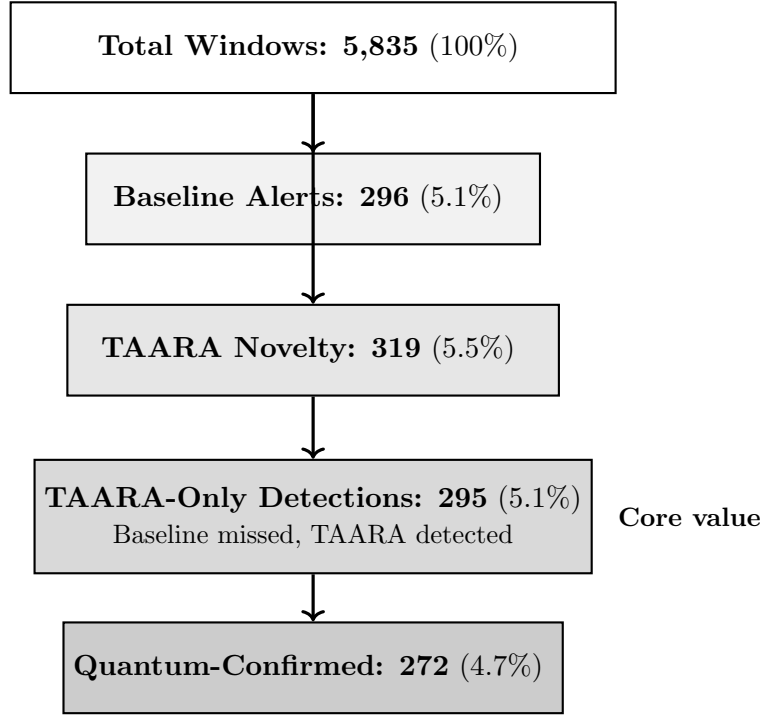


Figure 4: Detection funnel showing TAARA identifies 295 novel behavioral states missed by ensemble baseline methods, with 92.2% quantum-validated as directionally novel.

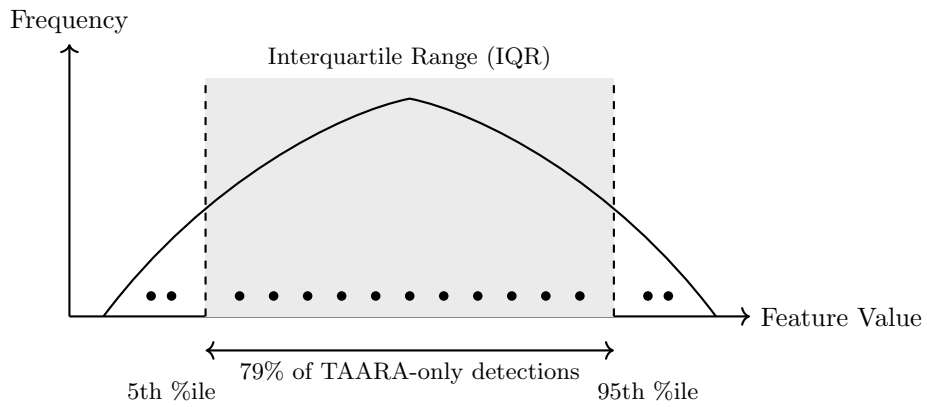


Figure 5: Statistical position of TAARA-only detections across feature distributions. 79% fall within the interquartile range, demonstrating that novelty detection is independent of statistical extremity.

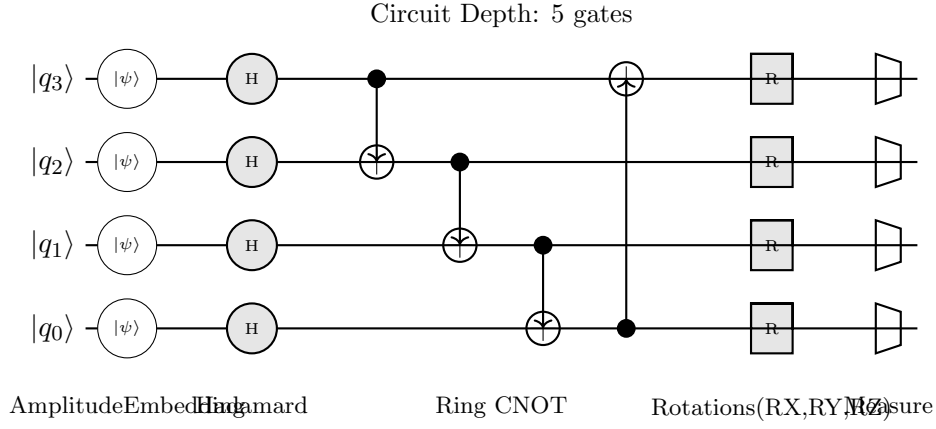


Figure 6: Quantum validation circuit for residual direction encoding. 4-qubit amplitude embedding with ring entanglement topology enables efficient fidelity measurement in Hilbert space.

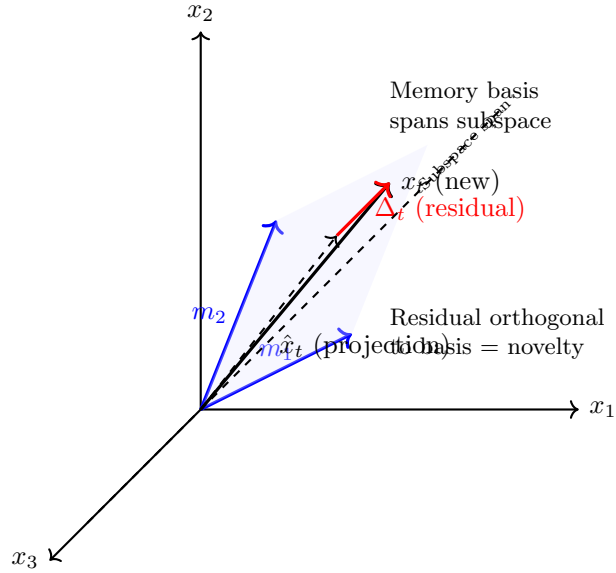
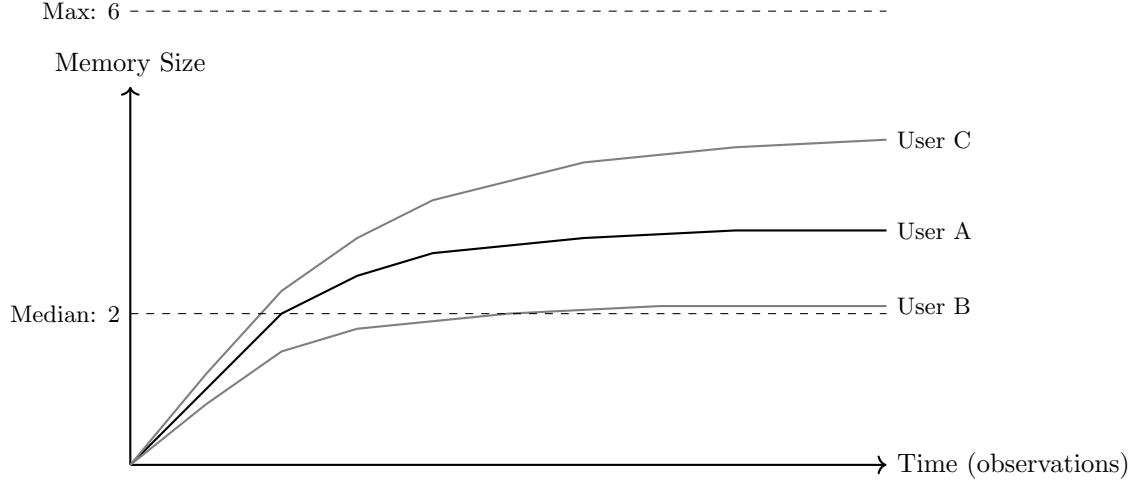


Figure 7: Reconstruction geometry (projected from 12D to 3D). New observation x_t projects onto memory basis span $\{m_1, m_2\}$. Residual Δ_t orthogonal to basis indicates behavioral novelty.



Memory bases stabilize at 2-6 states, not growing linearly

Figure 8: Memory efficiency over time for representative users. Per-identity memory bases stabilize at small sizes (median 2 states, max 6) rather than growing with observation count, demonstrating computational scalability.

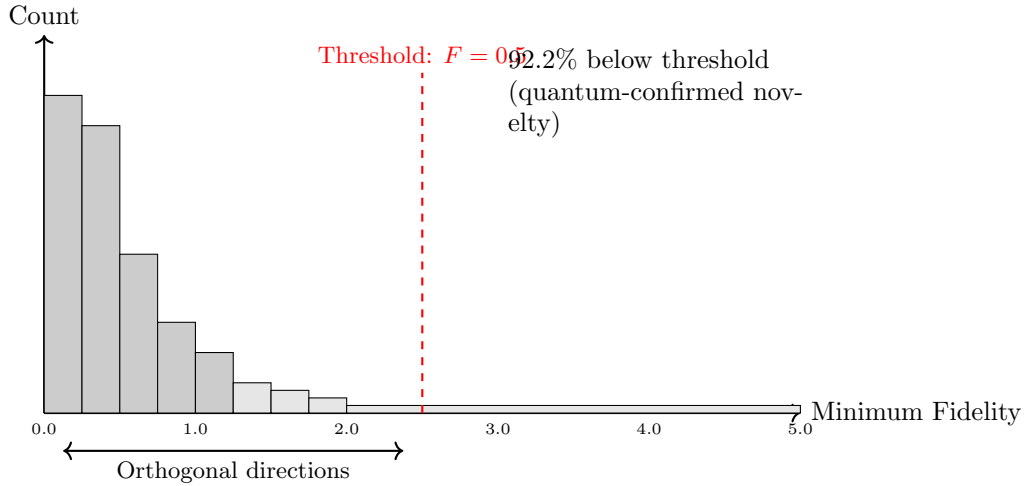


Figure 9: Quantum fidelity distribution for TAARA-only detections. Concentration near $F \approx 0$ confirms residual directions are orthogonal to memory bases, validating directional behavioral novelty rather than magnitude variation.